

Lyapunov Function Discovery: Transformer vs. Discrete Diffusion

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Side-by-side comparison of the Lyapunov functions discovered by the autoregressive transformer and the discrete-diffusion generator, on the systems where both arms have a finished seed. For each arm the *highest-reward* finished seed is shown. “Succeeded” is the training pipeline’s numerical check; “SMT” is the dReal verdict computed *identically for both arms* (same script, $\delta = 10^{-3}$, excluded ball radius 0.1, 300 s timeout). **strict**: $V > 0$ and $\dot{V} < 0$ proved; **LaSalle**: $V > 0$ and $\dot{V} \leq 0$ (asymptotic stability via the invariance principle); **not certified**: a counterexample was found (possibly a tolerance/boundary artifact); **timeout**: dReal did not finish. Caveat: the shown V is the *highest-reward* seed, which need not be the most certifiable one.

Simple pendulum (pendulum, dimension 2)

Dynamical system:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -0.1x_2 - \sin(x_1) \end{aligned}$$

Transformer: $V = x_2^2 - 2 \cos(x_1)$
reward $r = 0.9869$, succeeded: **Yes**, SMT: **LaSalle** ($V > 0$, $\dot{V} \leq 0$).

Diffusion: $V = x_2^2 - 2 \cos(x_1)$
reward $r = 0.9788$, succeeded: **Yes**, SMT: **LaSalle** ($V > 0$, $\dot{V} \leq 0$).

Van der Pol oscillator (vdp2d, dimension 2)

Dynamical system:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_1 - x_2(1 - x_1^2) \end{aligned}$$

Transformer: $V = x_2^2 - \cos(x_1) - \cos(2x_1 + x_2)$
reward $r = 0.9888$, succeeded: **Yes**, SMT: **strict** ($V > 0$, $\dot{V} < 0$).

Diffusion: $V = x_1^2 + x_2^2$
reward $r = 0.9376$, succeeded: **Yes**, SMT: **not certified**.

2-D polynomial system (poly2d, dimension 2)

Dynamical system:

$$\begin{aligned}\dot{x}_1 &= -5x_1^3 - 2x_1x_2^2 \\ \dot{x}_2 &= -9x_1^4 + 3x_1^3x_2 - 4x_2^3\end{aligned}$$

Transformer: $V = 9x_1^2 - \cos(2x_2)$

reward $r = 0.9921$, succeeded: **Yes**, SMT: **not run**.

Diffusion: *no finished seed yet (still running).*

3-D trigonometric system (trig3d, dimension 3)

Dynamical system:

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -x_2 - \sin(x_1)\cos(x_1) - \sin(x_3)\cos(x_3) \\ \dot{x}_3 &= x_2 - x_3\end{aligned}$$

Transformer: $V = x_2^2 + x_3^2 - \cos(2x_1 + x_2)$

reward $r = 0.9981$, succeeded: **Yes**, SMT: **strict** ($V > 0$, $\dot{V} < 0$).

Diffusion: $V = x_2^2 + x_3^2 + \sin^2(x_1 + x_2)$

reward $r = 0.9954$, succeeded: **Yes**, SMT: **not certified**.

3-D polynomial system A (poly3dA, dimension 3)

Dynamical system:

$$\begin{aligned}\dot{x}_1 &= -3x_1^3 + 3x_1x_3 - 9x_1 \\ \dot{x}_2 &= -x_1^3 - 5x_2 + 5x_3^2 \\ \dot{x}_3 &= -9x_3^3\end{aligned}$$

Transformer: $V = x_1^2 + x_2^2 - \cos(x_2 + x_3)$

reward $r = 0.9986$, succeeded: **Yes**, SMT: **not run**.

Diffusion: *no finished seed yet (still running).*

3-D polynomial system B (poly3dB, dimension 3)

Dynamical system:

$$\begin{aligned}\dot{x}_1 &= -8x_1x_2^2 - 10x_2^4 \\ \dot{x}_2 &= -8x_2^3 + 3x_2^2 - 8x_2 \\ \dot{x}_3 &= -x_3\end{aligned}$$

Transformer: $V = -x_2 - (\cos(x_2) + \cos(x_3))^{16} + \sin(x_1^2)$

reward $r = 1.0000$, succeeded: **Yes**, SMT: **not run**.

Diffusion: *no finished seed yet (still running).*

2-bus lossy power system (power4d, dimension 4)

Dynamical system:

$$\dot{x}_1 = 0.5x_3 - 0.5x_4$$

$$\dot{x}_2 = -0.5x_3 + 0.5x_4$$

$$\dot{x}_3 = -1.0x_3 - 0.5 \sin(x_1 - x_2) + 0.5 \cos(x_1 - x_2) - 0.5$$

$$\dot{x}_4 = -1.0x_4 + 0.5 \sin(x_1 - x_2) + 0.5 \cos(x_1 - x_2) - 0.5$$

Transformer: $V = x_3^2 + 2x_4^2 + (-x_1 + x_2 - x_3)^2$

reward $r = 0.9973$, succeeded: **Yes**, SMT: **dReal timeout**.

Diffusion: $V = x_4^2 - \cos(x_3) - \cos(-x_1 + x_2 + x_4)$

reward $r = 0.9903$, succeeded: **Yes**, SMT: **dReal timeout**.

6-D polynomial system (poly6d, dimension 6)

Dynamical system:

$$\dot{x}_1 = -x_1 + 0.5x_2 - 0.1x_5^2$$

$$\dot{x}_2 = -0.5x_1 - x_2$$

$$\dot{x}_3 = -0.1x_1^2 - x_3 + 0.5x_4$$

$$\dot{x}_4 = -0.5x_3 - x_4$$

$$\dot{x}_5 = -x_5 + 0.5x_6$$

$$\dot{x}_6 = 0.1x_2^2 - 0.5x_5 - x_6$$

Transformer: $V = x_6^2 - (\cos^4(x_2) + \cos(x_3x_4x_6 + x_5)) \cos(x_1)$

reward $r = 0.9988$, succeeded: **Yes**, SMT: **not certified**.

Diffusion: $V = x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 + (-x_1 + x_6)^2$

reward $r = 0.9987$, succeeded: **Yes**, SMT: **strict** ($V > 0$, $\dot{V} < 0$).

Quadrotor (quad6d, dimension 6)

Dynamical system:

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -1.0x_1 - 2.5x_2 - 1.5x_4x_6 - 0.5x_4 \sin(x_2) \cos(x_4) \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= 1.5x_2x_6 + 0.5x_2 \sin(x_2) \cos(x_4) - 1.0x_3 - 10.0x_4 \\ \dot{x}_5 &= x_6 \\ \dot{x}_6 &= -1.0x_5 - 0.8x_6\end{aligned}$$

Transformer: $V = x_1^2 + x_3^2 + x_4^2 + x_5^2 + x_6^2 + (x_1 + x_2)^2$
reward $r = 0.9982$, succeeded: **Yes**, SMT: **not certified**.

Diffusion: $V = (x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 + x_6^2)^8$
reward $r = 0.9849$, succeeded: **Yes**, SMT: **LaSalle** ($V > 0$, $\dot{V} \leq 0$).

8-D polynomial system (poly8d, dimension 8)

Dynamical system:

$$\begin{aligned}\dot{x}_1 &= -x_1 + 0.5x_2 - 0.1x_5^2 \\ \dot{x}_2 &= -0.5x_1 - x_2 \\ \dot{x}_3 &= -0.1x_1^2 - x_3 + 0.5x_4 \\ \dot{x}_4 &= -0.5x_3 - x_4 \\ \dot{x}_5 &= -x_5 + 0.5x_6 + 0.1x_7^2 \\ \dot{x}_6 &= -0.5x_5 - x_6 \\ \dot{x}_7 &= -x_7 + 0.5x_8 \\ \dot{x}_8 &= -0.1x_4^2 - 0.5x_7 - x_8\end{aligned}$$

Transformer: $V = x_1^2 + x_2^2 + x_3^2 + x_5^2 + x_6^2 + x_7^2 + x_8 \sin(x_8) - \cos(x_4)$
reward $r = 0.9995$, succeeded: **Yes**, SMT: **not run**.

Diffusion: *no finished seed yet (still running).*

9-D synthetic system (syn9d, dimension 9)

Dynamical system:

$$\begin{aligned}\dot{x}_1 &= -x_1 + 0.5x_2 - 0.1x_5^2 \\ \dot{x}_2 &= -0.5x_1 - x_2 + 0.1x_8 \\ \dot{x}_3 &= -0.1x_1^2 - x_3 + 0.5x_4 \\ \dot{x}_4 &= -0.5x_3 - x_4 \\ \dot{x}_5 &= -x_5 + 0.5x_6 \\ \dot{x}_6 &= 0.1x_2^2 - 0.5x_5 - x_6 \\ \dot{x}_7 &= x_8 \\ \dot{x}_8 &= -0.1x_2 - x_8 - \sin(x_7) \cos(x_7) - \sin(x_9) \cos(x_9) \\ \dot{x}_9 &= x_8 - x_9\end{aligned}$$

Transformer: $V = x_2^2 + x_3^2 + x_4^2 + x_5^2 + x_6^2 + x_8^2 + x_9^2 + (-x_1 + x_6)^2 + \sin^2(x_7)$
reward $r = 0.9992$, succeeded: **Yes**, SMT: **not certified**.

Diffusion: $V = x_3^2 + x_5^2 + x_8^2 + (x_1^2 + 4x_4^2 + x_6^2 + x_9^2)^2 - \cos(x_2) - \cos(x_7)$
reward $r = 0.9977$, succeeded: **No**, SMT: **dReal timeout**.

10-D polynomial system (poly10d, dimension 10)

Dynamical system:

$$\begin{aligned}\dot{x}_1 &= -x_1 + 0.5x_2 - 0.1x_5^2 \\ \dot{x}_2 &= -0.5x_1 - x_2 \\ \dot{x}_3 &= -0.1x_1^2 - x_3 + 0.5x_4 \\ \dot{x}_4 &= -0.5x_3 - x_4 \\ \dot{x}_5 &= -x_5 + 0.5x_6 + 0.1x_9^2 \\ \dot{x}_6 &= -0.5x_5 - x_6 \\ \dot{x}_7 &= -x_7 + 0.5x_8 \\ \dot{x}_8 &= -0.5x_7 - x_8 \\ \dot{x}_9 &= 0.5x_{10} - x_9 \\ \dot{x}_{10} &= -x_{10} - 0.1x_4^2 - 0.5x_9\end{aligned}$$

Transformer: $V = x_1^2 + x_{10}^2 + x_3^2 + x_5^2 + x_7^2 + (x_2 - x_4 + x_6 + x_8 + x_9)^2 + \sin(x_4^2)$
reward $r = 0.9993$, succeeded: **No**, SMT: **not run**.

Diffusion: *no finished seed yet (still running).*

Lossless power system (lossless6d, dimension 6)

Dynamical system:

$$\dot{x}_1 = x_4$$

$$\dot{x}_2 = x_5$$

$$\dot{x}_3 = x_6$$

$$\dot{x}_4 = -x_4 - 0.5 \sin(x_1 - x_2) - 0.5 \sin(x_1 - x_3)$$

$$\dot{x}_5 = -x_5 + 0.5 \sin(x_1 - x_2) - 0.5 \sin(x_2 - x_3)$$

$$\dot{x}_6 = -x_6 + 0.5 \sin(x_1 - x_3) + 0.5 \sin(x_2 - x_3)$$

Transformer: $V = x_4^2 + x_5^2 + x_6^2 + (-x_1 + x_2)^2 + (-x_1 + x_3 - x_4)^2$
reward $r = 0.9973$, succeeded: **No**, SMT: **not run**.

Diffusion: *no finished seed yet (still running).*